

RESEARCH ARTICLE

Ecological determinants of avian malaria infections: An integrative analysis at landscape, mosquito and vertebrate community levels

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Abstract

1. Vector and host communities, as well as habitat characteristics, may have important but different impacts on the prevalence, richness and evenness of vector-borne parasites.
2. We investigated the relative importance of (1) the mosquito community composition, (2) the vertebrate community composition and (3) landscape characteristics on the prevalence, richness and evenness of avian *Plasmodium*. We hypothesized that parasite prevalence will be more affected by vector-related parameters, while host parameters should be also important to explain *Plasmodium* richness and evenness.
3. We sampled 2,588 wild house sparrows (*Passer domesticus*) and 340,829 mosquitoes, and we performed vertebrate censuses at 45 localities in the Southwest of Spain. These localities included urban, rural and natural landscapes that were characterized by several habitat variables.
4. Twelve *Plasmodium* lineages were identified in house sparrows corresponding to three major clades. Variation partitioning showed that landscape characteristics explained the highest fraction of variation in all response variables (21.0%–44.8%). *Plasmodium* prevalence was in addition explained by vector-related variables (5.4%) and its interaction with landscape (10.2%). Parasite richness and evenness were mostly explained by vertebrate community-related variables.
5. The structuring role of landscape characteristics in vector and host communities was a key factor in determining parasite prevalence, richness and evenness, although the role of each factor differed according to the parasite parameters studied. These results show that the biotic and abiotic contexts are important to explain the transmission dynamics of mosquito-borne pathogens in the wild.

KEYWORDS

bird and mammal community, environmental predictors, mosquito community composition, NDVI, *Plasmodium*, vector-borne disease, water reservoir

1 | INTRODUCTION

The transmission dynamics of vector-borne parasites under natural conditions is determined by factors affecting the three main actors, the pathogen, its vectors and its vertebrate hosts, that are also influenced by environmental factors (Reisen, 2010). Currently, most studies have only focused on the study of pairwise interactions between two of the three main groups, while researches integrating all three groups were scarce and their relative contributions to the transmission of blood parasites remain unclear. Avian *Plasmodium* is a widespread vector-borne genus of parasites that infect birds and are transmitted by mosquitoes, mainly those of the genus *Culex* (Valkiūnas, 2005). These parasites represent an excellent bird model for human malaria and other vector-borne pathogens to investigate the ecological determinants of their transmission dynamics under natural conditions (Clark, Clegg, & Lima, 2014).

The abundance, species composition and diversity of vectors may affect pathogen amplification and ultimately determine the prevalence of the pathogen in the host population (Manguin & Boëte, 2011; Martínez-de la Puente, Martínez, Rivero-de-Aguilar, Del Cerro, & Merino, 2013a; Roche, Rohani, Dobson, & Guégan, 2013; Sol, Jovani, & Torres, 2000). This could be due to multiple demographic factors, such as alterations of the ratio between insect vectors and reservoirs and the vector-host contact patterns that potentially affect pathogen amplification (Joseph, Mihaljevic, Orlofske, & Paull, 2013; Roche et al., 2013). Different studies in birds have found strong support for the positive relationship between blood parasite prevalence and the abundance of potential insect vectors (Martínez-de la Puente et al., 2013a; Okanga, Cumming, & Hockey, 2013; Sol et al., 2000). In addition, differential vector competence of mosquito species may determine the rate of parasite transmission through the specific relationships between insect species and the pathogens infecting birds (Beerntsen, James, & Christensen, 2000; Martínez-de la Puente, Martínez, Aguilar, Herrero, & Merino, 2011), as only a subset of the parasites that contact with potential insect vectors are effectively transmitted by them (Gutiérrez-López et al., 2016). However, with some exceptions (Gager, Del Rosario Loaiza, Dearborn, & Bermingham, 2008), most evidence suggests that species of avian *Plasmodium* are generalist parasites transmitted by mosquitoes of different genera (Ferraguti et al., 2013; Kimura, Darbro, & Harrington, 2010). Furthermore, the feeding preferences of insect vectors may determine the contact rates between infected and susceptible hosts, affecting the transmission dynamics of mosquito-borne pathogens (Takken & Verhulst, 2013). Although some mosquito species show an opportunistic behaviour, they usually prefer to feed on particular vertebrate taxa, as those feeding mainly on birds (i.e. ornithophilic species) or on mammals (i.e. mammophilic species) (Takken & Verhulst, 2013). Thus, the abundance and diversity of mosquitoes, especially bird biters, may determine the transmission dynamics of *Plasmodium* parasites circulating in the area and affect parasite prevalence (Roche et al., 2013). The mosquito community might, however, have a lower impact on parasite

diversity and richness due to the low specificity of mosquitoes to transmit avian *Plasmodium* (Kimura et al., 2010).

The composition of vertebrate host communities may also affect the dynamics of parasites circulating in the area, due to host composition can considerably influence disease risk (Huang, Van Langevelde, Estrada-Peña, Suzán, & De Boer, 2016; Joseph et al., 2013; Kilpatrick, Daszak, Jones, Marra, & Kramer, 2006; Scordato & Kardish, 2014). In the last decades, the protecting effect of biodiversity on human health from pathogen transmission is generating an intense debate (Ostfeld & Keesing, 2012). However, most studies report an effect of host richness (the number of different species in an area) and not diversity (relative abundance of each species in an area) on pathogen prevalence, suggesting a key role of the composition structure of the host communities on the disease amplification (LoGiudice et al., 2008). Positive correlations have been reported between host abundance and richness and parasite richness (Johnson, Ostfeld, & Keesing, 2015; Kamiya, O'Dwyer, Nakagawa, & Poulin, 2014; Lafferty, 2012), including studies on different host-parasite models (Hechinger & Lafferty, 2005; Krasnov, Shenbrot, Khokhlova, & Degen, 2004; Watters, 1992). In the case of avian *Plasmodium*, parasite prevalence increases with the density (Ortego & Cordero, 2010) and abundance of susceptible hosts (Drovetski et al., 2014; but see Zhang, Wu, Zhang, Su, & Zou, 2014). The *Plasmodium* genus contains generalist and cosmopolitan lineages, which are able to infect bird species belonging to different families (Clark et al., 2014), but there are also some lineages that infect only a restricted range of species (Moens & Perez-Tris, 2016). Thus, the bird species present in a community may drive the blood-parasite community circulating in the area (Loiseau et al., 2012), resulting in similar parasite communities in areas with similar avian communities (Fecchio et al., 2017). In fact, the diversity of *Plasmodium* lineages may reflect the composition of avian hosts cohabiting in the area and be more affected by the bird host community than the mosquito community (Ishtiaq et al., 2010). Finally, although the presence of mammal host may elevate mosquito density and rise malaria risk at regional level (Dobson et al., 2006), an increase in reservoir species richness leads to a greater dilution capacity because the mammal community can accommodate possibilities of mosquito wasted bites, yielding overall lower infection prevalence for avian pathogens (Roche et al., 2013). Thus, mammals may divert bites in the short term but increase mosquito abundance in the long term (Bouma & Rowland, 1995).

Environmental characteristics may also determine the dynamics of parasite transmission (Higgs & Beaty, 2005; Sehgal, 2015) and incidence of vector-borne pathogens (Estrada-Peña, Ostfeld, Peterson, Poulin, & de la Fuente, 2014; Fourcade, Keišs, Richardson, & Secondi, 2014; Patz, Graczyk, Geller, & Vittor, 2000; Sehgal, 2015), probably through their effects on vector and host populations (Estrada-Peña et al., 2014; Roche et al., 2013). A number of studies have highlighted the importance of factors such as vegetation cover (Lillesand, Kiefer, & Chipman, 1994; Reisen, Meyer, Tempelis, & Spoehel, 1990; Roiz, Ruiz, Soriguer, & Figuerola, 2015) and water sources (Ferraguti et al., 2016; Ishtiaq et al., 2008; Krams et al., 2010; Njabo et al., 2009) on both vector and host communities. Moreover, water availability was

shown to be one of the most important predictors of avian malaria incidence in wild birds (Krama et al., 2015; Wood et al., 2007).

The role of environmental factors has been shown to play a particularly prominent role in driving the transmission dynamics of vector-borne pathogens such as malaria parasites (Balls et al., 2004; Foley, Torres, Mueller, Bryan, & Bell, 2003), including in forcing the infection dynamics of different avian *Plasmodium* lineages (Lachish, Knowles, Alves, Wood, & Sheldon, 2011). However, whether and to what extent landscape characteristics affect the prevalence of vector-borne parasites either directly (Sehgal, 2015) or indirectly through effects on hosts (Cardon, Loot, Grenouillet, & Blanchet, 2011) and/or vectors (Santiago-Alarcón, Palinauskas, & Schaefer, 2012) remains unknown. In addition, understanding the ecology of pathogens in urban environments and their consequences to hosts living in habitats of different quality is essential to manage the risk of transmission to wildlife, especially under the current human-driven environmental changes (Bradley & Altizer, 2007; Cornet, Bichet, Larcombe, Faivre, & Sorci, 2014). Differences in habitat type may influence parasite epidemiology, with an increase in the transmission of pathogens in urban areas (Bradley & Altizer, 2007). In fact, human-mediated landscape transformation may influence the dynamics of wildlife diseases by affecting the distribution of vectors (Ferraguti et al., 2016; Gilioli & Mariani, 2011), hosts (Bradley & Altizer, 2007), and consequently their contact rates and feeding ratio (Faraji et al., 2014; Valerio et al., 2010). Among others, some evidence supports the role of urban environments as amplifiers of the impact of avian malaria infections on hosts. For instance, habitat alterations and fragmentation (Shochat, Warren, Faeth, McIntyre, & Hope, 2006), as well as urbanization and alteration of trophic dynamics (Faeth, Warren, Shochat, & Marussich, 2005), affect the risk of pathogen transmission. Also, the high trace of metals concentrations is linked to the prevalence of avian malaria parasites and affects body condition of urban birds (Bichet et al., 2014), and a higher parasite virulence has been found in birds feeding on poor quantity and quality food from urban areas (Cornet et al., 2014). Finally, animal husbandry or agriculture intensification can also affect the transmission of vector-borne pathogens, including avian malaria parasites (Fourcade et al., 2014; Gonzalez-Quevedo, Davies, & Richardson, 2014; Patz et al., 2000). Thus, anthropogenic changes in landscapes and their subsequent effects on host communities can affect disease emergence in different situations (Jones et al., 2008).

Here, we used an integrative approach to disentangle the roles of the different actors: (1) mosquito abundance and community diversity, (2) vertebrate density and community diversity and (3) environmental characteristics and landscape use in mediating the prevalence, richness and evenness of *Plasmodium* parasites in wild populations of house sparrows (*Passer domesticus*). The house sparrow is an abundant and widespread species that occurs in a wide range of habitats, usually linked to human constructions, and has a highly variable prevalence of infection by *Plasmodium* in different populations (Marzal et al., 2011). Despite its adaptation to human inhabited areas, house sparrow populations are declining in many European cities (De Laet & Summers-Smith, 2007). We predicted

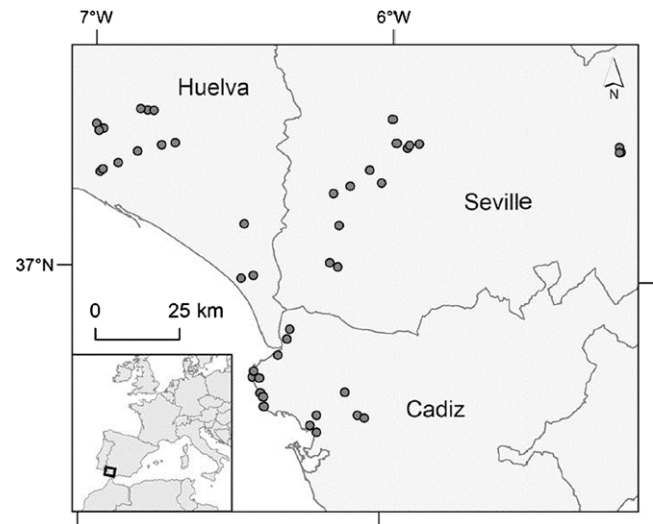


FIGURE 1 Distribution of the 45 sampling sites in the Southwest of Spain. The localities were distributed in geographically nearby triplets formed by one natural, one rural and one urban site

that *Plasmodium* prevalence is determined to a greater extent by mosquito-related variables (Roche et al., 2013), while *Plasmodium* richness and diversity will be more influenced by host-related variables (Lacorte et al., 2013). Habitat characteristics were predicted to globally affect *Plasmodium* transmission in birds due to their role in structuring both mosquito (Ferraguti et al., 2016) and vertebrate communities (Fecchio et al., 2017).

2 | MATERIALS AND METHODS

This study was conducted during 2013 in southwest Spain, at 45 different localities in the Cadiz, Huelva and Seville provinces (15 localities in each province; Figure 1). The study area is characterized by a Mediterranean climate, with a long dry summer season and most precipitation concentrated in autumn/winter. In each locality, we sampled three nearby areas (triplets) differing in anthropogenic use, defined as natural, rural and urban habitats. Urban habitats contained more densely populated areas than the other two habitat types; rural habitats had a higher density of livestock than the other two habitat types; and natural habitats were selected on the basis of both lower human and livestock densities than in the other two habitat types and of less human degraded landscape. The selection of these habitat categories was based on visual inspection and land-use geographical information. Vertebrate counts and mosquito and bird captures were done at each of these 45 localities. Three teams were simultaneously involved in house sparrow sampling, while mosquito trapping and vertebrate censuses were conducted during two working weeks and repeated every 45 days in all localities. All habitats within the same localities were visited during the same day to sample mosquitoes and perform the vertebrate censuses, and over consecutive days to sample birds.

2.1 | Mosquito sampling and identification

Mosquitoes were captured from April to December with BG-sentinel traps baited with BG-lure and dry ice as a source of CO₂ following the procedure detailed by Ferraguti et al. (2016). Briefly, mosquito sampling was conducted by using three traps operating for 24 hours at each sampling site. Mosquitoes were sorted by gender and date of collection, and then morphologically identified to species level (see Ferraguti et al., 2016 for further details). We quantified the number of mosquitoes captured as well as the number of ornithophilic and mammophilic mosquito species, based on available information about the feeding preferences of the different species: *Culex pipiens*, *Cx. modestus* and *Cx. perexiguus* feed mainly on birds (Balenghien et al., 2008; Muñoz et al., 2012), while *Cx. theileri*, *Anopheles atroparvus* and *Ochlerotatus caspius* feed mainly on mammals (Balenghien et al., 2008; Martínez-de la Puente, Ruiz, Soriguer, & Figuerola, 2013b; Martínez-de la Puente et al., 2012).

2.2 | Bird sampling and vertebrate censuses

House sparrows were captured at the same localities where mosquitoes were sampled after the breeding season, from July to October. Birds were captured using mist-nets, ringed with numbered metal rings (Svensson, 2006) and immediately released at the place of capture. A blood sample was obtained from the jugular vein using sterile syringes. The volume of blood extracted depended on the bird size but never exceeded 1% of the bird body mass. Blood was collected in Eppendorf tubes, maintained in cold boxes in the field and stored at 4°C for 24 hr prior to centrifugation for 10 min at 1,700 × g (4,000 rpm) to separate serum and cellular fractions. Samples were frozen at -20°C until subsequent molecular analysis.

Vertebrate censuses were conducted bimonthly from June to November, with a mean of 3.8 censuses per locality. Five point-counts in the surroundings of the mosquito traps were used for estimating bird abundance (Buckland, 2006). Although the vertebrate community could vary between seasons (i.e. due to the arrival of migrant individuals), we included the mean value of vertebrate censuses conducted during the whole study period in order to cover all the bird captures and because the *Plasmodium* community can be affected by the arrival of such migrant individuals through the year. A trained observer recorded the species and distance to the observer of all detected birds, both visually and acoustically, during 6 minutes in each sampling point. The density of each bird species at each locality was estimated by point transect analysis using the program DISTANCE (Thomas et al., 2006). For the mammal censuses, we counted pellets in transects of 200 m length and 1 m wide in the neighbouring area of the mosquito traps. Detected mammals included hare, rabbit, horse, cattle, red deer, fallow deer, wild boar, cat, dog and fox. Published information on the defecation rates of each species was used to estimate species densities (see Table S1 and Appendix S1).

2.3 | Molecular analyses

Genomic DNA was isolated from the cell fraction of each blood sample using the Maxwell®16 LEV system Research (Promega, Madison, WI) (Gutiérrez-López, Martínez-de la Puente, Gangoso, Soriguer, & Figuerola, 2015a). *Plasmodium* parasites were identified according to Hellgren, Bensch, and Malmqvist (2008) based on the amplification of a 478 bp fragment (excluding PCR primers) of the mitochondrial cytochrome *b* (cyt *b*) gene. Molecular analyses of negative samples were repeated to avoid false negatives (McClintock et al., 2010). Both negative controls for PCR reactions (at least one per plate) and DNA extraction (one per 15 samples) were included in the analysis. Positive amplifications were sequenced using the Macrogen sequencing service (Macrogen Inc., The Netherlands). PCR amplifications were sequenced uni-directionally from the 5' end using the primer HaemF to reduce the costs of sequencing the high number of amplicons (see Dubiec et al., 2016 for a similar procedure). However, the identity of each lineage was confirmed by sequencing amplicons bi-directionally with the complementary primer. Multiple infections were found in five samples, as we observed two or more lineages simultaneously supported by the presence of double peaks in the chromatogram, thus we only identified parasites at the genus level in these samples. Despite the existence of methods to resolve double lineages (Pérez-Tris & Bensch, 2005), these procedures were not used due to the small number of coinfections. Sequences were edited using the software Sequencher™ v.4.9 (Gene Codes Corp., © 1991–2009, Ann Arbor, MI, USA) and identified by assigning unknown cyt *b* sequences to previously identified parasite lineages recorded in the GenBank DNA sequence and MalAvi databases (Bensch, Hellgren, & Pérez-Tris, 2009). Novel lineages were classified following the recommendations of Bensch et al. (2009) and deposited in GenBank.

2.4 | Phylogenetic analyses of *Plasmodium* lineages

We used MEGA6 (Tamura, Stecher, Peterson, Filipski, & Kumar, 2013) to perform a phylogeny reconstruction including the complete dataset of 31 *Plasmodium* lineages with known morphospecies available from MalAvi (Bensch et al., 2009) and the sequences from the 12 lineages found in this study (including two lineages previously characterized morphologically as *P. relictum*: SGS1 and GRW11). Phylogenetic analyses were carried out based on the maximum likelihood algorithm. Analyses were conducted using the Tamura-Nei model and nodal support was estimated by bootstrap analysis with 10,000 replications (Felsenstein, 1981). One sequence of *Leucocytozoon majoris* (lineage LCB1, AY393804.1) was used as out-group. The lineages found infecting house sparrows in this study were grouped into three major clusters of phylogenetically related lineages, including sequences corresponding to *P. relictum*, *P. cathemerium* and *P. rouxi* morphospecies (see Section 3).

For each sampling site, we calculated the (1) *Plasmodium* prevalence, (2) richness of the three major clades found, named here on as *P. relictum* clade, *P. cathemerium* clade and *P. rouxi* clade, and

(3) evenness of clades, measured using the Shannon's equitability index (Agrawal & Gopal, 2013) (Table S2a).

Evenness was calculated as the Shannon's equitability index based on the prevalence of each *Plasmodium* clade at each locality. Shannon equitability index takes into account both species abundance and evenness.

2.5 | Landscape characteristics and remote sensing variables

Ten different landscape variables were estimated from satellite images (Table S2b). The hydrological information was obtained from cartography accessible at the "Datos Espaciales de Referencia de Andalucía (DERA)" webpage: <http://www.juntadeandalucia.es/institutodeestadisticaycartografia/DERA/>. We measured the distance from each mosquito trap to the nearest river, marshland patch, stretch of freshwater, coast line and main water reservoirs following the procedure detailed in Ferraguti et al. (2016). We also calculated

the minimum distance to any kind of water sources. The Normalized Difference Vegetation Index (NDVI) was estimated from MODIS sensor images at a spatial resolution of 250 m and a temporal resolution (MOD13Q1 product) of 16 days. See Ferraguti et al. (2016) for further details.

2.6 | Statistical analyses

Statistical analyses were conducted with R version 2.14.2 using the *vegan*, *lme4*, *car*, *arm*, *Matrix*, *Rcpp*, *MASS* and *MuMIn* packages. We performed Linear Mixed-Effects Models (LMM) fitted by maximum likelihood with Gaussian distribution, including *Plasmodium* prevalence, richness or evenness as dependent variables. In order to enhance the statistical power of our analyses, we compared the three habitat categories (natural, rural and urban areas) within the same triplet to account for any geographical stratification of our sampling design that may influence parasite/host/vector distribution or abundance. Thus, *triplet* variable was nested within *province* variable as

TABLE 1 Estimate (est), *z*, *p* and values from the LMMs of each *Plasmodium* variable (columns) and the predictors (rows) included in the study (*N* = 45 sampling localities). Independent variables have been grouped into different categories. AOS, abundance of ornithophilic species; AMS, abundance of mammophilic species; *E_H*, Evenness (see the complete description of all predictors in Table S2). Significant associations (*p* < .05) are shown in bold

	Prevalence			Richness of clades			Evenness		
	est	<i>z</i>	<i>p</i>	est	<i>z</i>	<i>p</i>	est	<i>z</i>	<i>p</i>
Mosquito community									
AOS				0.35	2.48	.01			
AMS	11.55	2.47	.01	−0.23	1.43	.15			
Richness							−0.09	0.96	.32
Mosquito <i>E_H</i>	−7.93	1.94	.04				−0.20	2.42	.02
Vertebrate community									
Sparrows sampled	−5.51	1.21	.23	0.26	1.92	.04	0.09	0.93	.34
Sparrows density	−7.02	1.66	.09						
Bird density	−4.15	0.95	.34						
Bird richness				0.21	1.42	.16	−0.09	0.94	.33
Bird <i>E_H</i>	5.12	1.22	.23	0.27	1.97	.05			
Mammal density									
Mammal richness									
Mammal <i>E_H</i>	−5.76	1.37	.17						
Landscape									
Rural									
Urban									
Coast	8.01	1.68	.08	0.50	3.47	<.001	0.48	6.05	<.001
Rivers	6.12	1.44	.14	0.28	2.24	.01	0.12	1.65	.10
Water reservoirs	10.55	2.01	.03				0.35	3.75	<.001
Freshwater	−7.50	1.57	.11	0.17	1.43	.15			
Marshland	−1.62	2.43	.02	0.15	1.19	.22			
Total water				0.25	1.81	.07	0.20	2.58	.01
Spring NDVI	9.93	2.21	.03	0.22	1.70	.09			
Summer NDVI				0.20	1.72	.09			

random factors in any model. *Provinces* were included in the model to control for large-scale geographical variation but also to control for the logistical structure of sample collection. We checked for collinearity in independent variables by calculating their variance inflation factor (VIF) in full and final models. The covariates with the highest VIFs were excluded sequentially until all VIF values were lower than 5 (Zuur, Ieno, & Elphick, 2010). First, we compared *Plasmodium* variables across the three habitat types using one-way ANOVAs to check for any habitat type influence. Then, LMMs were estimated to model the role of the three sets of explanatory variables: variables characterizing the mosquito community, the vertebrate community and the landscape (Table S2b). To find the best (final) models for each of the explanatory sets, we fitted different submodels with all the possible combinations of explanatory variables (within each set) and selected that with the lowest Akaike's information criterion (AIC) among those with $\Delta AIC \leq 2$. All explanatory variables present in these selected models were kept in the final model. The normal distribution of all predictors and model residuals was checked by using *qq plots* in R software. The mean abundance of ornithophilic and mammophilic mosquito species and hydrologic distances were log-transformed to normalize their distributions.

Finally, we used a variation partitioning approach to determine the proportion of variation explained by the exclusive and shared contributions of the three groups of explanatory variables (mosquitoes, vertebrates and landscape) for each response variable (*Plasmodium* prevalence, richness and evenness). These groups were formed by the selected variables according to the model selection procedure described earlier (Table 1). The habitat category was included in the set of landscape variables. To calculate the variation explained by each set of explanatory variables and their combinations, we calculated the respective marginal coefficient of determination (a pseudo- R^2) for the fixed effects according to Nakagawa and Schielzeth (2013), using the R package *MUMIN* (Bartoń, 2015). Subsequently, exclusive and shared fractions of explained variation were determined following the procedure described in Legendre (2008).

3 | RESULTS

A total of 340,829 female mosquitoes were collected, of which the most trapped species were *Cx. theileri* (282,891 ind.), *Oc. caspius* (21,155) and *Cx. pipiens* (19,268). In addition, *Cx. perexiguus* (5,939), *An. atroparvus* (5,387) and *Culex modestus* (1,237) reached high abundances in the study area.

The overall prevalence of infection by *Plasmodium* was 29.6% (765 of 2,588 house sparrows) comprising 12 different lineages, five of them identified for the first time in this study. The two most common parasite lineages were SGS1 (GenBank accession number KR049254; prevalence 67.5%) and COLL1 (=Donana07) (AY831747, prevalence 13.5%). The phylogenetic analysis of the *Plasmodium* sequences found in this study revealed that these were grouped in three main clades with previously morphologically characterized

Plasmodium lineages (Figure 2). The lineages DELURB4 (EU154346), GRW11 (=Rinshi-7) (KR049255), SGS1, PADOM25 (KX438373), PADOM26 (KX438374) and PADOM28 (KX438376) were grouped in the *P. relictum* clade with a 99% bootstrap support, while COLL1, PADOM02 (=Rinshi-8) (AB477127), PADOM1 (DQ058611), PADOM27 (KX438375) and PADOM29 (KX438378) were included in the *P. cathemerium* clade with a 63% bootstrap support and the lineage PAGRI02 (JX196865) clustered with one lineage corresponding to *P. rouxi* with a 99% bootstrap support. Parasites of the *P. relictum* clade (23.22%) showed the highest prevalence of infection followed by those of the *P. cathemerium* clade (5.99%) and *P. rouxi* clade (0.15%).

We did not find any significant difference on *Plasmodium* prevalence or richness between habitat categories (ANOVA: all $p > 0.05$). A marginal significant difference was found between habitat categories for *Plasmodium* evenness ($\chi^2 = 5.2$, $df = 2$, $p = .06$), with rural habitat showing higher values than natural areas (Tukey's post hoc test: $z = 2.31$, $p = .05$).

As predicted, the effect of mosquito, vertebrate and landscape predictors varied among the different *Plasmodium* indexes. *Plasmodium* prevalence was positively associated with the abundance of mammophilic mosquitoes, the distance to water reservoirs and spring NDVI, but decreased with the evenness of mosquitoes and distance to marshlands (Table 1). No relationships were identified between *Plasmodium* prevalence and the vertebrate community variables.

Plasmodium richness significantly increased with the abundance of ornithophilic mosquitoes, evenness of bird community, total number of sparrows sampled and distance to the coast and rivers. The relationships between *Plasmodium* richness and distance to any source of water and both NDVI indexes were marginally significant (Table 1).

Plasmodium evenness was negatively related to the mosquito evenness and positively related to the distance to the coast, water reservoirs and to any sources of water. No significant relationship was found between *Plasmodium* evenness and the vertebrate community variables (Table 1).

The global models, which included the three sets of explanatory variables, explained a total variation in *Plasmodium* prevalence, richness and evenness ranging from 46.2% to 60.0% (Figure 3). Overall, landscape characteristics exclusively explained the highest fraction of variation (21.0%–44.8%). For the case of *Plasmodium* prevalence, the mosquito and vertebrate communities explained similar fractions of exclusive variation (5.4% and 5.5%, respectively), and the shared fraction between landscape and the mosquito community explained a large proportion of the variation (10.2%) (Figure 3a). Both the mosquito and vertebrate communities explained variation in parasite richness, although the vertebrate community explained a larger fraction (3.5% compared with 1.1%) (Figure 3b). *Plasmodium* evenness was also largely explained by the exclusive contribution of the mosquito community (6.8%), whereas the vertebrate community explained a limited amount of variation (2.5%) (Figure 3c,d). Explained variation shared between landscape and the mosquito

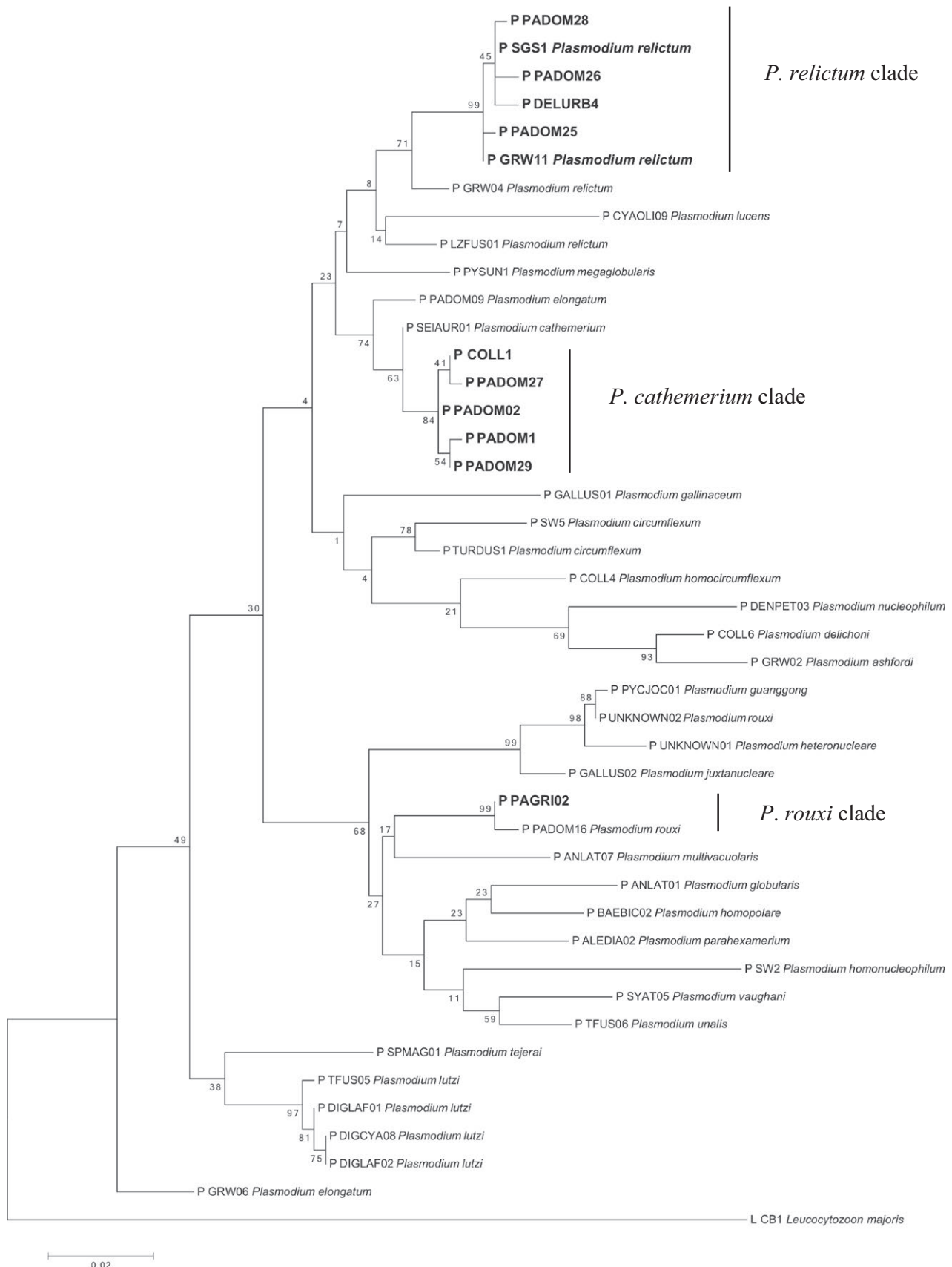


FIGURE 2 Phylogenetic tree of the lineages isolated from house sparrows in this study (in bold) and the *Plasmodium* lineages corresponding to morphospecies available in MalAvi database (Bensch et al., 2009). The three main clades were named according to the morphospecies included in each cluster

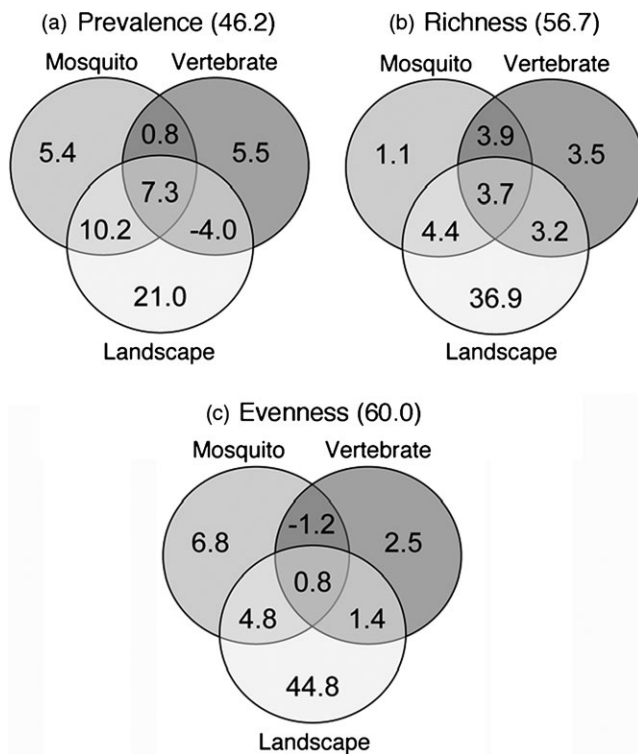


FIGURE 3 Venn diagrams of the variation partitioning results for the *Plasmodium* prevalence (a), richness (b) and evenness (c). All fractions are given in percentage (%). The global coefficient of variation (R^2) is provided in each panel title within brackets

community was higher than that between landscape and the vertebrate community for parasite prevalence, richness and evenness.

4 | DISCUSSION

Parasite infections are known to vary in space and time, and several abiotic or biotic parameters are likely responsible for this variation (Higgs & Beaty, 2005). Our large-scale survey was based on an unprecedented analysis of the role of vectors, hosts and environment in determining parasite prevalence, richness and evenness. This integrative approach allowed us to quantify their relative importance in explaining parasite biodiversity. Overall, our results support that landscape characteristics and their role in structuring the vector and host communities is the most likely factor driving *Plasmodium* biodiversity. The mosquito community also explained a large proportion of variation in *Plasmodium* prevalence, while the vertebrate community was more important in explaining parasite richness.

Landscape features explained most of the variation in all *Plasmodium* variables, both exclusively and through their role in structuring the mosquito and vertebrate communities. The exclusive contribution of landscape might also affect mosquitoes and vertebrates in ways that are not captured by the variables measured in this study. For example, habitat variables can affect their community composition and relative abundances and in turn explain

parasite dynamics. Thus, environmental factors such as the distance to water sources and NDVI, reflecting characteristics such as vegetation cover and phenology, have been shown to be good predictors in the transmission dynamics of pathogens (Foley et al., 2003; Omumbo, Hay, Snow, Tatem, & Rogers, 2005; Sehgal, 2015). Strong relationships between landscape use and the prevalence of blood parasites infecting birds have previously been shown (Fourcade et al., 2014; Patz et al., 2000), which may be due to the effect of landscape features on the vector community (Norris, 2004). For instance, the scarcity of suitable vectors has been identified as the most likely reason of the low prevalence or absence of *Plasmodium* in birds from marine and saline environments (Figuerola, 1999; Gutiérrez-López et al., 2015b; Martínez-de la Puente et al., 2017; Mendes, Piersma, Lecoq, Spaans, & Ricklefs, 2005) and deserts (Merino, Barbosa, Moreno, & Potti, 1997). Indeed, environmental characteristics strongly determined the mosquito abundance and community composition in the study area (Ferraguti et al., 2016). In particular, we found positive relationships between *Plasmodium* prevalence, richness and evenness with the distance to different manmade water reservoirs, and a negative association between the distance to the marshland and *Plasmodium* prevalence. As salt-marshes provide suitable environments for halophytic mosquito species (Leishnam & Sandoval-Mohapatra, 2011), marshland could strongly affect the abundance of some mosquitoes in the studied areas (Ferraguti et al., 2016). Moreover, in addition to mosquito breeding, it is also possible that the landscape affects the mosquito host-seeking behaviour potentially affecting the dynamics of parasite transmission. Here, these results agree with previous studies reporting an increased probability of *Plasmodium* infection in the proximity of freshwater, ponds and slow rivers in mosquitoes (Ishtiaq et al., 2008; Njabo et al., 2009) and birds (Krams et al., 2010; Wood et al., 2007). In addition, the environmental conditions reflected by higher NDVI values may be related to the availability of breeding and resting habitats for both mosquitoes and birds (Lillesand et al., 1994; Reisen et al., 1990). In fact, the NDVI has been related to the spatial distribution of mosquitoes (Diuk-Wasser, Brown, Andreadis, & Fish, 2006; Roiz et al., 2015). Therefore, the positive association between vegetation cover and mosquito presence, abundance and diversity may explain the associations between spring NDVI and *Plasmodium* prevalence reported here. Also, previous studies reported a peak in prevalence during spring in the bird breeding season, when vector populations and the proportion of immunologically naive juveniles in the host population are high and higher levels of sexual steroid hormones depress the immune system, allowing parasites to survive (Saino, Møller, & Bolzern, 1995; Wedekind & Følstad, 1994). This fact is supported by the high shared contribution (10.2%) of the mosquito community and landscape characteristics to explain *Plasmodium* prevalence, which supports the role of the environment in structuring the vector community and contributes to explain parasite prevalence in house sparrows. However, this association should be also due to the human transformation of the habitat owing to the creation of artificial habitats such as water deposits, swimming pools or urban sewage that act as alternative breeding

sites for mosquitoes (Bradley & Altizer, 2007; Overgaard, Ekbom, Suwonkerd, & Takagi, 2003).

The mosquito community alone also explained most of the variance in *Plasmodium* prevalence, supporting that vector abundance and diversity are determinants of the prevalence of vector-borne parasites in birds (Martínez-de la Puente et al., 2013a; Sol et al., 2000). The abundance of mosquitoes could affect avian parasite prevalence (Manguin & Boëte, 2011), especially the ornithophilic species that are responsible for successful transmission (Simpson et al., 2012; Takken & Verhulst, 2013). By identifying ornithophilic mosquitoes involved in the transmission of *Plasmodium* parasites (Ventim et al., 2012), we predicted an increase in *Plasmodium* prevalence in areas with a higher abundance of ornithophilic species (Kimura et al., 2010; Santiago-Alarcón et al., 2012), while a significant effect was not necessarily expected for the case of mammophilic mosquitoes. We found a positive association between the abundance of ornithophilic mosquitoes and *Plasmodium* richness but, contrary to our prediction, the prevalence of *Plasmodium* was positively correlated with the abundance of mammophilic mosquito species. Although the feeding preference of these mosquitoes is clearly biased towards mammals, this is not a strict behaviour, with a proportion of individuals feeding also on birds and interacting with avian blood parasites (Ferraguti et al., 2013). Moreover, under altered environmental conditions, vectors specialized in a group of vertebrate hosts can be forced to feed on other groups of organisms because of changes in host community structure (Lyimo & Ferguson, 2009). For example, the mammophilic *Oc. caspius* include birds in their diet, with species including chickens and house sparrows representing 18.1% and 9.1% of the blood meals of mosquitoes trapped in two areas of southern Spain (Muñoz et al., 2012). In addition, these species are potentially involved in the transmission of *Plasmodium* (Ferraguti et al., 2013), which together with the high abundance recorded in the studied area suggest that mammophilic mosquitoes may still favour the transmission of avian *Plasmodium* to birds. Finally, we found negative relationships between mosquito evenness and both *Plasmodium* prevalence and diversity. This fact could be due to a differential competence of mosquitoes cohabiting in the area, which may play a differential role in the transmission of avian *Plasmodium* with respect to their relative abundance. Furthermore, according to previous molecular studies, *Plasmodium* lineages are often generalist in terms of their vectors (Clark et al., 2014; Ferraguti et al., 2013; Kimura et al., 2010; Tolle, 2009 but see Gager et al., 2008). These studies are, however, based on the presence of parasite DNA, but potential quantitative interspecific differences in parasite loads are currently unknown. Quantitative differences in vector competence for different *Plasmodium* lineages should be studied in future to identify the causes of the observed pattern.

The vertebrate community also contributed to explain parasite prevalence and diversity. It is widely accepted that pathogen diversity is spatially correlated with host biodiversity (Lafferty, 2012; Wood et al., 2014), wherein an increase in host diversity may be linked with dilution or amplification of diseases (Huang, Yu, Van Langevelde, & De Boer, 2017; Ostfeld & Keesing, 2012; Randolph & Dobson, 2012). As predicted, we found *Plasmodium* richness

positively related with bird evenness, supporting an increase in the number of parasite clades circulating in house sparrows as bird richness and diversity increased in the area. Thus, contrary to the dilution effect predictions, *Plasmodium* parasites were never negatively associated to host richness nor diversity, suggesting the occurrence of an amplification rather than a dilution effect. Moreover, Ishtiaq et al. (2010) found that the richness of bird species affected parasite richness more than the species richness of mosquitoes. Likewise, bird diversity was correlated to parasite diversity, supporting the hypothesis that high host richness might provide more niches to the different parasite lineages (Lacorte et al., 2013). Our results not only support this possibility but also that even abundances may favour the richness of *Plasmodium* lineages infecting birds.

Surprisingly, we did not find support for the urbanization effect on *Plasmodium* variables, which does not support a differential exposure hypothesis of these parasites, as previously found (Bichet et al., 2014). While in some cases animals living in urban habitats may be protected from infectious diseases, there have been reports of increased transmission of pathogens in urban areas (Bradley & Altizer, 2007). At this point, it is worth noting the dual role of anthropization on host communities and the incidence of infectious diseases. Indeed, urbanization can drastically affect wildlife populations due to the high amount of pollutants that might weaken the avian immune system and increase their susceptibility to infectious diseases (Fair & Myers, 2002; Hawley, Hallinger, & Cristol, 2009; Snoeijs et al., 2005). Nevertheless, these same urbanization factors can also reduce parasite prevalence in birds from urban habitats (Evans et al., 2009; Geue & Partecke, 2008).

In spite of the large sample size included in this study, our results are limited by the fact that *Plasmodium* was only analysed in a single host. Avian *Plasmodium* lineages may vary in their ability to infect different species (Palinauskas, Valkiūnas, Bolshakov, & Bensch, 2008; Zhang et al., 2014), and some parasite morphospecies are thought to be host restricted (Iezhova, Valkiūnas, & Bairlein, 2005). The two most abundant lineages found in this study, belonging to different clades, were already described in 116 (SGS1) and 10 (COLL1) different host species (according to MalAvi; accessed 27 November 2017, see Bensch et al., 2009), supporting that these lineages are generalist and not restricted to house sparrows, possibly affecting the bird community of the study areas. However, our approach allowed us to deal with the problem of differential detection associated with heterogeneous sample size among host species (see Loiseau et al., 2012), although the latter approach limits the isolated lineages to those from a single host species.

To conclude, we present evidence that biotic and abiotic factors play an important role in *Plasmodium* transmission. We show that it is essential to develop integrative studies, focusing in all the players affecting the dynamics of vector-borne diseases to gain a more comprehensive understanding of the ecology of vector-borne pathogens in natural ecosystems. While vectors and hosts are both necessary links in parasite transmission dynamics, the environmental context plays a determinant role in regulating the regional incidence and diversity of pathogens.

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AUTHORS' CONTRIBUTIONS

M.F., J.M.P., R.S. and J.F. conceived and designed the experiments; M.F., J.M.P., D.R. and S.R. collected samples and performed the experiments; M.F., J.M.P., S.B., D.V. and J.F. carried out the data analysis; J.F. and R.S. contributed to the reagents/materials/analysis tools. M.F., J.M.P., S.B., D.V., D.R., S.R., R.S. and J.F. wrote the paper.

DATA ACCESSIBILITY

The data associated with new lineage information described in this manuscript are accessible from GenBank (KX438373-KX438378).

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